SHENZHEN MSU-BIT UNIVERSITY

Department of Chemistry

«Course: Environmental Chemistry»

[Your Name]

Review on Chemical Processes Across the Atmosphere, Hydrosphere and Lithosphere and Their Mutual Feedback Mechanisms with Global Climate Change

Course Assignment

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Abstract

Global climate change is, at its core, the macroscopic expression of a planetary network of chemical reactions distributed across the atmosphere, hydrosphere and lithosphere and coupled through the biosphere. This review synthesizes the present scientific understanding of the dominant chemical processes operating within each of these three geospheres and, critically, of the mutual feedback mechanisms that link their chemistry to the trajectory of global climate. In the atmosphere we examine greenhouse-gas chemistry and lifetimes, the hydroxyl-radical-controlled oxidative capacity, aerosol microphysics and aerosol–cloud–radiation interactions, and stratospheric–tropospheric ozone chemistry. In the hydrosphere we review the ocean carbon sink and carbonate buffering, ocean acidification, the biological carbon pump, deoxygenation, and the cycling of nutrients and dissolved gases. In the lithosphere we treat the silicate-weathering "thermostat," carbonate and sulfide weathering, volcanic and metamorphic outgassing, mineral-associated soil organic carbon, permafrost carbon, and engineered enhanced rock weathering. We then integrate these processes through the coupled global biogeochemical cycles of carbon, nitrogen, sulfur, phosphorus and iron, and finally assess the spectrum of climate feedbacks—from fast physical feedbacks and climate sensitivity, through carbon-cycle and methane feedbacks, to abrupt tipping elements and the ultimate geological negative feedback. We emphasize the asymmetry of timescales: the stabilizing silicate-weathering feedback operates over ~10^5 years and cannot buffer anthropogenic emissions, whereas several amplifying feedbacks act on decadal-to-centennial timescales relevant to twenty-first-century policy. Quantitative estimates, principal uncertainties, and open research questions are highlighted throughout.

Keywords: biogeochemical cycles; carbon cycle; ocean acidification; silicate weathering; climate feedbacks; tipping points; atmospheric chemistry; Earth system science.

1. Introduction

The Earth functions as a coupled physical–chemical–biological system in which the chemistry of the gaseous, liquid and solid envelopes of the planet is continuously exchanging matter and energy. The Intergovernmental Panel on Climate Change concludes that it is unequivocal that human influence has warmed the climate, with a best estimate of equilibrium climate sensitivity of 3 °C and a transient climate response of 1.8 °C [1]. The proximate driver of this warming is a chemical perturbation: the injection of carbon dioxide, methane and nitrous oxide into the atmosphere at rates that natural sinks cannot match. In 2023 global fossil-fuel CO_2 emissions reached 10.1 ± 0.5 GtC yr^{-1}, and only about half of all anthropogenic CO_2 is removed each year by land and ocean sinks, with the remainder accumulating in the atmosphere [2]. Yet the perturbation is not confined to a single reservoir. Reactive nitrogen and phosphorus flows have been driven so far beyond their Holocene baselines that the biogeochemical-flows planetary boundary is now assessed as transgressed into a high-risk zone [3].

Understanding climate change therefore requires understanding chemistry across geosphere boundaries. The atmosphere, hydrosphere and lithosphere are not isolated reactors; they are linked by fluxes of carbon, nitrogen, sulfur, phosphorus, oxygen and trace metals, and these fluxes are themselves sensitive to climate. The recognition that a chemical process can stabilize climate over geological time dates to the foundational hypothesis that temperature-dependent silicate weathering provides a negative feedback maintaining Earth's surface temperature within habitable bounds [4]. Modern Earth-system science extends this insight into a dense web of feedbacks, some stabilizing and some amplifying, operating across a vast range of timescales.

This review is organized to mirror the structure of the Earth system itself. Sections 2–4 treat the chemistry internal to each geosphere—atmosphere, hydrosphere and lithosphere—in turn. Section 5 integrates these processes through the coupled global biogeochemical cycles that physically connect the spheres. Section 6 then analyzes the mutual feedback mechanisms linking this chemistry to global climate change, distinguishing fast physical feedbacks, carbon-cycle and methane feedbacks, and abrupt tipping elements. Section 7 synthesizes cross-cutting themes and identifies priorities for future research. Throughout, our aim is to make explicit both the direction (amplifying or stabilizing) and the characteristic timescale of each chemical–climate coupling, because it is the mismatch between these timescales that frames the central challenge of the Anthropocene.

2. Chemical Processes in the Atmosphere

2.1 Greenhouse-gas chemistry, lifetimes and sinks

The radiative forcing of the atmosphere is governed by the abundance and lifetime of its trace constituents. Carbon dioxide, having risen to roughly 50% above its 1750 level, is the single largest contributor to anthropogenic radiative forcing; the total anthropogenic effective radiative forcing over 1750–2019 is assessed at 2.72 (1.96 to 3.48) W m^{-2}, of which CO_2 alone contributes on the order of +2.16 W m^{-2} [5]. Unlike the other long-lived greenhouse gases, CO_2 has no single chemical sink in the atmosphere; its removal is governed by partitioning among the ocean and land reservoirs over a spectrum of timescales, so that a substantial fraction of an emission pulse persists for millennia.

Methane, by contrast, is removed chemically. Its total atmospheric lifetime is approximately 9.1 years, dominated by oxidation by the tropospheric hydroxyl radical, with smaller sinks from stratospheric loss, tropospheric halogen chemistry and soil uptake [5]. A subtlety of profound importance for climate accounting is that methane's *perturbation* lifetime exceeds its total atmospheric lifetime: an increase in methane emissions depresses tropospheric OH, which in turn lengthens methane's own lifetime, a positive CH_4–OH feedback that is encoded as a multiplicative factor in global-warming-potential calculations [5]. Reflecting this and carbon-cycle feedbacks, IPCC AR6 assigns methane a 100-year global warming potential of 27 (biogenic) to 29.8 (fossil) and a 20-year potential of roughly 81–83 [5]. For the 2008–2017 decade, total global methane emissions were approximately 576 Tg CH_4 yr^{-1}, of which about 359 Tg CH_4 yr^{-1} (~60%) were anthropogenic [7] (Fig. 3).

[Fig. 3 — paste figure here]

Fig. 3. The global methane budget: natural and anthropogenic CH_4 sources and the principal sinks. *Source: Saunois et al., The Global Methane Budget 2000-2017, Earth Syst. Sci. Data, 2020, 12: 1561 [7] (open access, CC BY 4.0).* Download: https://essd.copernicus.org/articles/12/1561/2020/

Nitrous oxide is the third major long-lived greenhouse gas and the dominant stratospheric-ozone-depleting emission of the twenty-first century. It has a mean atmospheric lifetime of 116 ± 9 years, with its principal sink being photolysis and $O(^1D)$ oxidation in the tropical middle stratosphere [8]. Atmospheric N_2O rose from a pre-industrial ~270 ppb to 332–336 ppb by 2019–2022, producing an effective radiative forcing of 0.21 ± 0.03 W m^{-2}, roughly one-tenth of the CO_2 forcing [5], [8]. Crucially, anthropogenic N_2O emissions increased 40% from 1980 to 2020, with agricultural fertilizer and manure responsible for about 74% of the anthropogenic source in the 2010s [8]—an explicit chemical bridge between the lithospheric–biospheric nitrogen cycle and atmospheric radiative forcing.

2.2 Oxidative capacity and the hydroxyl radical

The hydroxyl radical is the dominant tropospheric oxidant and effectively sets the chemical lifetime of methane, carbon monoxide and a host of reactive species; the global oxidizing capacity of the atmosphere is conventionally defined by the tropospheric mean OH concentration [9]. Because OH controls how quickly reactive greenhouse gases are destroyed, any climate- or emission-driven change in OH propagates into the greenhouse-gas burden. Multi-model analysis under the Aerosol Chemistry Model Intercomparison Project indicates that tropospheric OH changed little from 1850 to about 1980 and then increased by roughly 9% to 2014, shortening the methane lifetime; the global trend is dominated by the competing influences of nitrogen-oxide emissions (which raise OH) and methane and carbon-monoxide concentrations (which lower it) [10]. Observational constraints derived from the synthesis of airborne and satellite formaldehyde measurements reveal that bottom-up chemical models can carry significant biases in remote-troposphere OH, which in turn bias the inferred lifetimes of OH-removed species such as methane [11]. The oxidative capacity is thus both a chemical regulator of greenhouse-gas concentrations and a key, imperfectly constrained, term in climate–chemistry coupling.

2.3 Aerosols and aerosol–cloud–radiation interactions

Aerosols exert the largest negative—and most uncertain—anthropogenic radiative forcing. IPCC AR6 best estimates are −0.22 W m^{-2} for aerosol–radiation interactions and −0.84 W m^{-2} for aerosol–cloud interactions, the latter (5–95% range −1.43 to −0.29 W m^{-2}) constituting the single largest source of uncertainty in total anthropogenic forcing [5]. The chemistry of aerosol formation is therefore central to climate. Sulfate aerosol, formed by the oxidation of sulfur dioxide and dimethyl sulfide, scatters solar radiation and serves as cloud condensation nuclei. Black carbon, by contrast, is strongly absorbing: its industrial-era forcing through direct absorption, cloud effects and snow/ice-albedo reduction is estimated at +1.1 W m^{-2} (90% bounds +0.17 to +2.1 W m^{-2}), making it a leading warming agent after CO_2 [12]. Carbonaceous aerosol chemistry adds further complexity: top-down estimates place global secondary organic aerosol production at 140–910 TgC yr^{-1}, an order of magnitude above bottom-up accounting in which biogenic precursors (~88 TgC yr^{-1}) dominate over anthropogenic ones (~10 TgC yr^{-1}) [13]. Because secondary organic aerosol is a major fraction of submicron organic mass yet carries large uncertainty in its direct and indirect forcing, it remains a leading research challenge for quantifying aerosol–climate coupling [14] (Fig. 2).

[Fig. 2 — paste figure here]

Fig. 2. Effective radiative forcing of the climate system by component over 1750-2019. *Source: IPCC AR6 WGI, Chapter 7, Fig. 7.6 (Forster et al.) [5]*. Download: https://www.ipcc.ch/report/ar6/wg1/figures/chapter-7/figure-7-6/

2.4 Ozone chemistry and stratosphere–climate coupling

Ozone is simultaneously a greenhouse gas in the troposphere and a protective ultraviolet shield in the stratosphere, and its chemistry couples tightly to climate. The tropospheric ozone burden has risen approximately 45% since 1850, and the combined tropospheric-plus-stratospheric ozone change contributes an effective radiative forcing of +0.47 (0.24 to 0.71) W m^{-2}, with the dominant uncertainty stemming from pre-industrial ozone-precursor (nitrogen-oxide, carbon-monoxide, volatile-organic-compound) emissions [5]. In the stratosphere, halogen chemistry destroying ozone has been brought under control by the Montreal Protocol: under continued compliance, ozone is projected to recover to 1980 values by about 2066 over Antarctica, 2045 over the Arctic, and 2040 elsewhere [15]. Because the controlled ozone-depleting substances are also potent greenhouse gases, their phase-out is estimated to avoid roughly 0.5 °C of additional warming by mid-century, a striking example of direct coupling between stratospheric halogen chemistry and the radiative climate [15].

3. Chemical Processes in the Hydrosphere

3.1 The ocean carbon sink and carbonate buffering

The ocean is the largest reactive carbon reservoir in rapid exchange with the atmosphere, and it has absorbed a major fraction of anthropogenic CO_2. Inventory reconstructions establish that the ocean accumulated 118 ± 19 PgC of anthropogenic carbon between roughly 1800 and 1994 [16], and a further 34 ± 4 PgC between 1994 and 2007, equal to 31 ± 4% of anthropogenic emissions over that interval [17]. Contemporary budgets place the ocean sink near 2.9 PgC yr^{-1}, removing roughly a quarter (about 26% over the 2014–2023 decade) of total anthropogenic emissions [2]. The chemistry underlying this uptake is the carbonate buffer system: dissolved CO_2 reacts with water and carbonate ion to form bicarbonate, so that the ocean's capacity to absorb additional CO_2 is governed by the Revelle factor. As dissolved inorganic carbon accumulates, the Revelle factor rises and a given carbon addition produces a disproportionately larger increase in seawater pCO_2, progressively weakening the chemical buffering and the efficiency of the solubility pump as the century proceeds [18] (Fig. 4).

[Fig. 4 — paste figure here]

Fig. 4. Change in the column inventory of anthropogenic carbon stored in the ocean. *Source: Gruber et al., Science, 2019, 363: 1193 [17] (copyright AAAS; access via publisher/institution).* Download: https://www.science.org/doi/10.1126/science.aau5153

3.2 Ocean acidification

The same chemistry that sequesters CO_2 acidifies the ocean. Surface-ocean pH has fallen by about 0.1 units since pre-industrial times, from roughly 8.2 to 8.1, corresponding to a greater than 30% increase in hydrogen-ion concentration, at a recent rate of approximately 0.017–0.027 pH units per decade [19]. Anthropogenic CO_2 penetration has shoaled the aragonite and calcite saturation horizons by roughly 30–200 m relative to pre-industrial, propagating acidification from the surface into the ocean interior [6]. The biological consequences are concentrated in carbonate-secreting organisms. Modeling first showed that, under a business-as-usual trajectory, Southern Ocean surface waters begin to become aragonite-undersaturated when atmospheric CO_2 reaches ~550 ppm, with undersaturation extending across the entire Southern Ocean south of 60°S by 2100, threatening pteropods and cold-water corals [20]. Experimental work confirms the vulnerability: under combined acidification and warming, larval mortality in the polar pteropod *Limacina helicina* rose to as much as 39%, accompanied by extensive shell malformation and dissolution [21] (Fig. 5).

[Fig. 5 — paste figure here]

Fig. 5. Ocean acidification: decline in surface-ocean pH and shoaling of the aragonite saturation horizon. *Source: IPCC AR6 WGI, Chapter 5, Fig. 5.21 (Canadell et al.) [6].* Download: https://www.ipcc.ch/report/ar6/wg1/figures/chapter-5/figure-5-21/

3.3 The biological carbon pump

Beyond the solubility pump, marine biology actively transports carbon to depth. Recent estimates based on multidecadal hydrographic data place total organic-carbon export at the base of the euphotic zone at approximately 15.0 ± 1.1 PgC yr^{-1}, of which about 11.1 ± 1.0 PgC yr^{-1} is sequestered for longer than three months, with roughly 81% attributable to sinking particles and vertically migrating zooplankton [22]. This biological pump maintains the vertical gradient of dissolved inorganic carbon and thereby keeps atmospheric CO_2 substantially lower than it would otherwise be; its sensitivity to warming, stratification and nutrient supply is a key uncertainty in projecting the future ocean sink.

3.4 Ocean deoxygenation

Warming and stratification are altering the ocean's dissolved-gas chemistry. The global ocean has lost about 2% of its dissolved-oxygen inventory since the 1960s, the open-ocean low-oxygen area has expanded by roughly 4.5 million km^2, the anoxic volume has approximately quadrupled, and more than 500 coastal low-oxygen sites are now documented; a further 3–4% global decline is projected by 2100 [23]. Multi-model projections show that twenty-first-century surface warming and pH decline are accompanied by substantial subsurface oxygen and productivity declines, with the largest oxygen and pH decreases in intermediate and mode waters [24]. Deoxygenation is itself chemically consequential because low-oxygen waters are the principal sites of nitrous-oxide production and of nitrogen loss through denitrification (Fig. 6).

[Fig. 6 — paste figure here]

Fig. 6. Global distribution of declining dissolved oxygen in the open ocean and coastal waters. *Source: Breitburg et al., Science, 2018, 359: eaam7240 [23] (copyright AAAS; access via publisher/institution).* Download: https://www.science.org/doi/10.1126/science.aam7240

3.5 Nutrient cycling and dissolved trace gases

The chemistry of marine productivity is governed by nutrient limitation. Iron limits phytoplankton growth and shapes community composition over a substantial fraction—on the order of 20–40%—of the world ocean, the high-nitrate, low-chlorophyll regions of the Southern Ocean and equatorial and subarctic Pacific, with iron–nitrogen co-limitation common in transitional zones [26]. The fixed-nitrogen budget is set by the balance between nitrogen fixation and denitrification; water-column denitrification is estimated at approximately 66 ± 6 TgN yr^{-1}, a lower rate than some earlier estimates that allows the marine fixed-nitrogen budget to be approximately balanced [25]. Low-oxygen zones are also potent sources of nitrous oxide: the global oceanic N_2O efflux is approximately 3.5 TgN yr^{-1} [8], produced mainly by nitrification and denitrification in low-oxygen waters, where oxygen-deficient zones account for more than half of the microbial source [27]. The Peruvian oxygen-minimum-zone upwelling alone emits ~0.2–0.9 TgN as N_2O per year, equivalent to 5–22% of prior global marine estimates [28]. Finally, although the marine gas-hydrate reservoir is enormous (of order 1,600–1,800 GtC), the authoritative assessment is that abrupt large-scale methane release this century is very unlikely, because only the small fraction of hydrate near the seafloor and close to the stability boundary is vulnerable, contributing at most tenths of a degree on millennial timescales [29]. The contemporary ocean carbon sink is, however, already responding to climate: warming-driven reductions in CO_2 solubility and increased stratification have lowered mean uptake by about 0.25 PgC yr^{-1} (~13% of the net flux) over 2000–2019 [30].

4. Chemical Processes in the Lithosphere

4.1 Silicate weathering and the geological thermostat

Over geological timescales, the chemistry of rock weathering regulates atmospheric CO_2. The chemical weathering of silicate minerals consumes CO_2—the cations released (principally Ca^2+ and Mg^2+) are ultimately reprecipitated as marine carbonate, sequestering carbon on timescales exceeding 10^5–10^6 years [33]. Because the weathering rate increases with temperature and runoff, it constitutes a temperature-dependent negative feedback: a warmer, higher-CO_2 climate accelerates weathering and draws CO_2 down, while a colder climate suppresses it [4]. This "thermostat" has maintained Earth's habitability despite a ~25–30% increase in solar luminosity over 4.6 billion years; quantitatively, at ~4.5 Ga with a Sun about 30% fainter, a CO_2 partial pressure of order 0.1 bar would have been required to keep the surface above freezing, pointing to the carbonate–silicate cycle as the resolving regulator of the faint-young-Sun paradox [32]. The effective global temperature sensitivity of silicate weathering is modest—an apparent activation energy of only about 22 kJ mol^{-1} when upscaled, because roughly half the land surface is runoff-limited or regolith-shielded and responds weakly to temperature [31]. Critically for the present, the e-folding response time of this negative feedback on atmospheric CO_2 is of order 10^5 years (reassessed at roughly 240,000 years), which is why it cannot buffer anthropogenic emissions on human timescales [34] (Fig. 7).

[Fig. 7 — paste figure here]

Fig. 7. The carbonate-silicate cycle and the silicate-weathering negative feedback (the long-term carbon thermostat). *Source: Penman et al., Earth-Sci. Rev., 2020, 209: 103298 [33] (copyright Elsevier; access via publisher/institution).* Download: https://doi.org/10.1016/j.earscirev.2020.103298

4.2 Carbonate, silicate and sulfide weathering: net carbon balance

Not all weathering is a long-term carbon sink, and the distinction is chemical. Carbonate weathering is rapid congruent dissolution, whereas silicate weathering proceeds by hydrolysis with secondary-clay precipitation; crucially, carbonate weathering does not provide a net long-term CO_2 sink because the carbon is returned to the atmosphere upon marine carbonate burial [36]. From the dissolved-load chemistry of the world's 60 largest rivers, global silicate weathering consumes atmospheric and soil CO_2 at roughly 8.7×10^{12} mol C yr^{-1} (~0.1 GtC yr^{-1}), establishing the present-day continental silicate sink [35]. The sign of the flux can even reverse: sulfuric acid generated by sulfide oxidation reacting with carbonate rocks is a net CO_2 *source* rather than a sink, so that the co-variation of silicate, carbonate and sulfide weathering in rapidly eroding mountain terrain can release CO_2 at rates about twice the sequestration rate of slowly eroding terrain on timescales shorter than marine sulfide compensation [37]. This complexity also underlies the long-standing uplift-weathering hypothesis, in which the monotonic late-Cenozoic rise in seawater $^{87}Sr/^{86}Sr$ was interpreted as evidence that Himalayan–Tibetan and Andean uplift enhanced silicate weathering, drew down CO_2, and drove the long global cooling into the Quaternary ice ages [38].

4.3 Volcanic and metamorphic outgassing

The solid-Earth source term that weathering must balance is volcanic and metamorphic CO_2 degassing. Field-based estimates of CO_2 output from modern volcanic arcs range from approximately 1.5–3.5 Tmol C yr^{-1}, with present-day contact-metamorphic decarbonation contributing a smaller ~0.06–0.9 Tmol yr^{-1} [39]. Direct measurement of non-eruptive degassing from Earth's most actively degassing subaerial volcanoes yields 51.3 ± 5.7 Tg CO_2 yr^{-1} over 2005–2015, with eruptive degassing adding only ~1.8 ± 0.9 Tg yr^{-1} [40]. On the multimillion-year timescale, the near-balance between this geological source and the silicate-weathering sink sets the baseline atmospheric CO_2 around which the thermostat operates.

4.4 Soil carbon and mineral–organic associations

The terrestrial lithosphere–biosphere interface stores vast quantities of carbon, and its stability is controlled by mineral chemistry. Soil organic matter is functionally partitioned into particulate organic matter and mineral-associated organic matter; the latter, dominated by microbial-derived, nitrogen-rich compounds that persist through chemical bonding to mineral surfaces, is more stable but saturates, and constitutes the dominant long-term soil-carbon reservoir [41]. Because mineral protection sets an upper bound on storage, the global stock of mineral-associated soil carbon (to ~1 m depth) has a quantifiable, mineralogically controlled saturation capacity that—though most soils remain well below it today—ultimately bounds how much additional carbon soils can sequester, an important constraint on natural-climate-solution potential [42].

4.5 Permafrost carbon

A distinctive lithospheric reservoir is the carbon frozen in permafrost. Northern permafrost-zone soils store approximately 1,460–1,600 PgC—about twice the current atmospheric carbon pool and roughly half of global soil organic carbon—and warming-driven thaw is projected to release on the order of 37–174 PgC as CO_2 and methane by 2100, constituting a positive climate feedback [46]. The most likely mode of release is gradual and sustained over decades to centuries rather than an abrupt "bomb," but it is nonetheless large enough to erode the remaining carbon budget consistent with the Paris Agreement temperature targets [47] (Fig. 8).

[Fig. 8 — paste figure here]

Fig. 8. Permafrost-zone carbon stocks and the permafrost carbon-climate feedback. *Source: Schuur et al., Nature, 2015, 520: 171 [46] (copyright Springer Nature; access via publisher/institution).* Download: https://www.nature.com/articles/nature14338

4.6 Enhanced rock weathering as engineered carbon removal

The same chemistry that regulates climate over geological time can be deliberately accelerated. Deploying basalt-based enhanced rock weathering on croplands could remove on the order of 0.5–2 Gt CO_2 yr^{-1} globally at extraction costs of roughly US$80–180 per tonne CO_2, with co-benefits for soil fertility and reduced ocean acidification [43]. A four-year maize–soybean field trial in the US Corn Belt applying crushed basalt achieved a conservative cumulative removal of 10.5 ± 3.8 t CO_2 ha^{-1} while raising crop yields by 12–16% [44]. Moreover, accounting for vegetation and soil responses to powdered-rock application can make the global removal potential substantially larger than estimates based on inorganic alkalinity chemistry alone [45]. Enhanced weathering thus represents an attempt to compress the geological thermostat's 10^5-year response onto a human-relevant timescale.

5. Cross-Sphere Coupling and Global Biogeochemical Cycles

The chemistry of the three geospheres is unified by global biogeochemical cycles that physically transport elements between them. These cycles are the connective tissue through which a perturbation in one sphere becomes a forcing on another.

5.1 The global carbon cycle

The carbon cycle is the master integrator. Anthropogenic CO_2 emissions are partitioned among the ocean sink (~23%), the land sink (~31%), and the atmosphere, with the airborne fraction averaging 44% over 1959–2019 despite large interannual variability [6]. This apparent stability conceals fragility. In 2023 the atmospheric CO_2 growth rate at Mauna Loa reached a record 3.37 ± 0.11 ppm—86% above 2022—even though fossil emissions rose only modestly, because the net land carbon sink collapsed to 0.44 ± 0.21 GtC yr^{-1}, its weakest since 2003, driven largely by a decline of the tropical-land sink under drought and heat [48]. Such episodes are direct, observable manifestations of the carbon–climate feedback discussed in Section 6, and they illustrate how the atmospheric burden is governed not only by emissions but by the climate sensitivity of the land and ocean reservoirs (Fig. 1).

[Fig. 1 — paste figure here]

Fig. 1. Global carbon budget: anthropogenic CO_2 sources and their partitioning among atmospheric growth, the ocean sink and the land sink. *Source: Friedlingstein et al., Global Carbon Budget 2024, Earth Syst. Sci. Data, 2025, 17: 965 [2] (open access, CC BY 4.0).* Download: https://essd.copernicus.org/articles/17/965/2025/

5.2 The nitrogen cycle and its human perturbation

Humanity has transformed the nitrogen cycle more profoundly than any other element cycle. Three industrial-era processes—Haber–Bosch fixation, cultivation of nitrogen-fixing crops, and fossil-fuel combustion—together now add more reactive nitrogen to the biosphere each year than all natural terrestrial processes combined [49]. Industrial Haber–Bosch fixation alone has grown to roughly 120 Tg N yr^{-1}, and an estimated two billion people depend on it for their food supply [50]. The consequences propagate across spheres: reactive-nitrogen inputs to the ocean via rivers and the atmosphere more than doubled from ~40 to ~90 Tg N yr^{-1} between pre-industrial times and the present [49], fueling coastal eutrophication and deoxygenation, while agricultural nitrogen drives the 74% of anthropogenic N_2O emissions noted in Section 2 [8]. The nitrogen cycle thus couples the lithosphere (fertilized soils), the hydrosphere (coastal hypoxia) and the atmosphere (N_2O forcing) in a single anthropogenic chain.

5.3 The sulfur cycle, dimethyl sulfide and the CLAW hypothesis

The sulfur cycle exemplifies both a proposed biological climate regulator and a major anthropogenic forcing. The CLAW hypothesis proposed a climate-regulating feedback in which marine phytoplankton emit dimethyl sulfide that oxidizes to sulfate aerosol—the dominant source of marine cloud condensation nuclei—thereby modulating cloud albedo and, in principle, surface temperature [51]. The natural global dimethyl-sulfide flux is approximately 28 Tg S yr^{-1} [53], roughly a third to a half of the anthropogenic atmospheric sulfur source [54]. However, subsequent observational analysis found the link between cloud condensation nuclei and cloud albedo too weak to support a substantial CLAW-type feedback, arguing against significant climate regulation via oceanic sulfur emissions [52]. Meanwhile, anthropogenic SO_2 emissions (~70–80 Tg S yr^{-1}, dominated by coal combustion) overtook natural sulfur sources around 1950 and now dominate the atmospheric sulfur budget, contributing the cooling sulfate aerosol forcing of Section 2 [54]. The sulfur cycle is therefore a vivid case in which a hypothesized negative feedback proved weak while the human perturbation became dominant.

5.4 The phosphorus cycle

Phosphorus, unlike carbon, nitrogen and sulfur, has no significant gaseous phase, so its cycle is dominated by lithospheric weathering and sedimentary burial—yet humans have perturbed it severely. Human mining of phosphorus (~23.5 Tg P yr^{-1}) now exceeds the pre-industrial weathering mobilization of 15–20 Tg P yr^{-1}, more than doubling the phosphorus flowing through the environment relative to the Holocene baseline [55]. Human activity has roughly tripled phosphorus mobilization in the land–water continuum, with net accumulation in soils of 6.9 ± 3.3 Tg P yr^{-1} [56]. Because phosphorus availability ultimately limits primary production on long timescales, this perturbation reshapes the productivity—and hence carbon uptake—of both terrestrial and aquatic ecosystems.

5.5 Iron, dust and the lithosphere–atmosphere–ocean linkage

Mineral dust is the physical vector that couples the solid Earth to the ocean and atmosphere. The iron hypothesis posits that dust-borne iron supply controls phytoplankton production over the high-nitrate, low-chlorophyll regions of the ocean that are iron-limited and thereby modulated glacial–interglacial CO_2 drawdown [57]. Dust also delivers phosphorus over intercontinental distances: roughly 27.7 Tg yr^{-1} of Saharan dust reaches the Amazon Basin, supplying ~0.022 Tg P yr^{-1}—an input comparable to the basin's hydrological phosphorus loss and thus essential to preventing long-term depletion [58].

Synthesizing such processes, recent assessment finds that mineral dust from the North African and Asian "dust belt" couples lithosphere, atmosphere and ocean by delivering iron and phosphorus that stimulate productivity and CO_2 drawdown, with total airborne phosphorus of order 1.4–3.5 Tg P yr^{-1} [59]. Dust is thus a tangible chemical conveyor belt linking weathering on land to fertility in the sea.

5.6 Planetary boundaries: an integrative diagnosis

The planetary-boundaries framework integrates these perturbations into a system-level diagnosis. As of its 2015 update, the biogeochemical-flows boundary for nitrogen and phosphorus was assessed as transgressed into the high-risk zone—one of four boundaries then exceeded, alongside climate change, biosphere integrity and land-system change [3]. The proposed safe boundary for nitrogen is roughly 62 Tg N yr^{-1} of intentional fixation, far below current rates, while for phosphorus it is approximately 11 Tg P yr^{-1} of flow to the ocean [3]. These transgressions underscore that the chemical coupling of the geospheres is no longer operating within its Holocene envelope.

6. Mutual Feedback Mechanisms with Global Climate Change

The chemical processes reviewed above do not merely respond to climate change; they feed back upon it. We organize the feedbacks by the timescale on which they act, from fast physical feedbacks to slow geological regulation.

6.1 Fast physical feedbacks and climate sensitivity

The amplitude of warming for a given forcing is set by the sum of fast feedbacks. The dominant negative feedback is the Planck (blackbody) response, assessed at approximately –3.3 W m^{-2} $°C^{-1}$ at Earth's mean surface temperature [5]. Acting against it, the combined water-vapor-plus-lapse-rate feedback is the largest positive contribution: the water-vapor component is ~1.77 ± 0.20 W m^{-2} $°C^{-1}$ in models (1.85 ± 0.32 from satellite observations), partially offset by a lapse-rate feedback of –0.50 ± 0.20 W m^{-2} $°C^{-1}$ [5]. The net cloud feedback is positive at +0.42 W m^{-2} $°C^{-1}$ (range –0.10 to +0.94), and a net negative cloud feedback is now considered very unlikely, though clouds remain the largest single source of feedback uncertainty [5]. Summing process-based estimates yields a net feedback parameter of –1.16 W m^{-2} $°C^{-1}$ (range –1.81 to –0.51)—stabilizing overall, yet permitting the assessed equilibrium climate sensitivity of 3 °C (likely 2.5–4 °C) and transient climate response of 1.8 °C [5], [1] (Fig. 9).

[Fig. 9 — paste figure here]

Fig. 9. Individual climate feedback parameters assessed by IPCC AR6. *Source: IPCC AR6 WGI, Chapter 7, Fig. 7.10 (Forster et al.) [5]*. Download: https://www.ipcc.ch/report/ar6/wg1/figures/chapter-7/figure-7-10/

6.2 The ice–albedo feedback

The cryosphere provides a powerful positive feedback whose sign is unambiguous. The loss of Arctic sea ice between 1979 and 2011 produced roughly +0.21 W m^{-2} of radiative forcing from albedo decline alone, equivalent to about one-quarter of the CO_2 forcing over the same interval [60]. More recently, Antarctic sea-ice loss has strengthened the global snow-and-ice albedo feedback, which exhibited a decadal trend of +0.08 ± 0.04 W m^{-2} per decade over 1992–2015 [61]. While not a chemical feedback per se, ice–albedo amplification warms the surface and ocean and thereby modulates the temperature-sensitive chemistry—solubility, weathering, methane release—treated throughout this review.

6.3 Carbon-cycle feedbacks

The carbon cycle couples to climate through two distinct feedbacks that must be carefully distinguished. The carbon–concentration feedback (β) describes the enhancement of land and ocean carbon uptake as atmospheric CO_2 rises; in CMIP6 models it is +0.97 ± 0.40 PgC ppm^{-1} over land and +0.79 ± 0.07 PgC ppm^{-1} over ocean—a negative, dampening feedback on climate [62]. The carbon–climate feedback (γ) describes the reduction in carbon uptake as the climate warms; it is –45.1 ± 50.6 PgC $°C^{-1}$ over land and –17.2 ± 5.0 PgC $°C^{-1}$ over ocean, a positive, amplifying feedback [62]. At present the dampening term dominates: natural sinks removed about 31% of emissions on land and 23% in the ocean over 2010–2019, offsetting more than half of emissions [6]. But IPCC AR6 assesses that, although absolute uptake continues to grow with rising CO_2, the *fraction* of emissions taken up declines as concentrations and temperatures rise—a net positive carbon–climate feedback whose early signature is visible in events such as the 2023 land-sink collapse [6], [48]. The permafrost feedback adds to this: AR6 estimates gradual thaw releases approximately 18 GtC per 1 °C of warming by 2100 (range 3–41 GtC $°C^{-1}$), and remaining carbon budgets are reduced by an additional 26 ± 97 $GtCO_2$ per °C to account for permafrost and other Earth-system feedbacks not captured by the core models [6].

6.4 The methane–wetland feedback

Methane biogeochemistry provides a further amplifying loop. Wetland methane emissions increase by approximately 24 ± 10 Tg CH_4 yr^{-1} per 1 °C rise in land-surface temperature, a positive methane–climate feedback [63]; under high-emission scenarios wetland emissions are projected to rise to 221–338 Tg CH_4 yr^{-1} by 2100 (from ~172 Tg yr^{-1} today), growing to dominate the global methane source and adding 0.04–0.19 W m^{-2} of radiative forcing [63]. Coupled with the CH_4–OH chemical feedback of Section 2, by which rising methane suppresses its own oxidant and lengthens its lifetime [5], the methane system contains nested positive feedbacks operating in both biology and chemistry.

6.5 Tipping points and abrupt change

Beyond gradual feedbacks lies the possibility of abrupt, potentially irreversible transitions. A systematic assessment of 16 climate tipping elements concludes that about five may be triggered at today's warming of ~1.1–1.5 °C—the Greenland and West Antarctic ice sheets, abrupt permafrost thaw, Labrador/subpolar-gyre convection collapse, and tropical coral-reef die-off—with more becoming likely above 1.5 °C; central thresholds include ~1.5 °C for the Greenland Ice Sheet and ~4 °C for collapse of the Atlantic Meridional Overturning Circulation [65]. The AMOC is a focus of active debate. IPCC AR6 assesses that it is very likely to weaken over the twenty-first century under all scenarios (projected ~30–50% weakening under high emissions) but that an abrupt collapse before 2100 is *not likely* with medium confidence [64]. In sharp contrast, an early-warning-signal analysis of observation-based AMOC fingerprints estimates a collapse central year of ~2057 (95% confidence interval 2025–2095), illustrating genuine and unresolved scientific disagreement about tipping timing [66]. Reassuringly, marine and terrestrial methane-hydrate dissociation is very unlikely to cause a detectable departure from emission trajectories this century, and IPCC AR6 does not include marine clathrates among its plausible near-term tipping points [29], [64] (Fig. 10).

[Fig. 10 — paste figure here]

Fig. 10. Global map of climate tipping elements and their estimated temperature thresholds. *Source: Armstrong McKay et al., Science, 2022, 377: eabn7950 [65] (copyright AAAS; access via publisher/institution).* Download: https://www.science.org/doi/10.1126/science.abn7950

6.6 The ultimate negative feedback and the timescale problem

Over the longest timescales, the silicate-weathering thermostat will eventually remove the anthropogenic carbon perturbation, restoring CO_2 toward its geologically buffered baseline [33], [4]. But its characteristic response time of order 170,000–380,000 years (mean ~240,000 years) renders it utterly irrelevant to the human predicament [34], [44]. Herein lies the central asymmetry of the Earth system's chemical feedbacks: the one robust, large negative feedback that has stabilized climate across deep time operates orders of magnitude too slowly to counter twenty-first-century warming, while several amplifying feedbacks—carbon–climate, permafrost, methane–wetland, ice–albedo—act on the decadal-to-centennial timescales that matter for policy. It is this mismatch, rather than any single process, that defines the chemical challenge of anthropogenic climate change.

7. Synthesis and Future Directions

Three cross-cutting themes emerge from this review. First, the geospheres are chemically inseparable: the nitrogen released by industrial fixation becomes atmospheric N_2O and coastal hypoxia; the sulfur burned from coal becomes cooling aerosol; the iron and phosphorus weathered from rock and lofted as dust fertilize the open ocean; and the carbon emitted from fossil reservoirs is partitioned, buffered, acidified and biologically pumped through every sphere. Any account of climate that treats these in isolation is incomplete. Second, the *sign* of a chemical–climate coupling is not fixed: silicate weathering is a sink while sulfide-driven carbonate weathering is a source; carbon-concentration feedbacks dampen warming while carbon-climate feedbacks amplify it; the hypothesized CLAW sulfur feedback proved weak while the anthropogenic sulfur perturbation proved strong. Third, and most consequentially, the feedbacks span an enormous range of timescales, and the protective negative feedbacks are concentrated at the slow end.

Several priorities for future research follow directly. The largest forcing uncertainty remains aerosol–cloud interactions, demanding improved constraints on secondary organic aerosol chemistry and on the microphysical link between aerosol composition and cloud albedo [5], [14]. The future trajectory of the ocean and land carbon sinks—and especially the magnitude of the carbon–climate feedback γ and the tropical-land sink revealed as fragile in 2023—requires sustained observation and better representation of nutrient limitation, drought and fire [62], [48]. The timing and reversibility of tipping elements, above all the AMOC, must be narrowed given the stark disagreement between process models and statistical early-warning analyses [64], [66]. The tropospheric oxidative capacity that governs methane's lifetime needs tighter observational constraint to reduce biases in greenhouse-gas accounting [10], [11]. And the real-world efficacy, permanence and co-benefits of engineered carbon removal through enhanced rock weathering require field validation at scale, as a deliberate attempt to recruit lithospheric chemistry against the anthropogenic perturbation [43], [44], [45].

8. Conclusions

The chemistry of the atmosphere, hydrosphere and lithosphere constitutes a single, deeply interconnected system whose perturbation by human activity is the proximate cause of global climate change, and whose feedbacks will substantially shape its course. We have shown that the atmospheric burden of greenhouse gases is set by chemical lifetimes and the hydroxyl-radical oxidative capacity; that the ocean both buffers and is acidified by the same carbonate chemistry, while warming erodes its uptake efficiency; that lithospheric weathering provides the planet's ultimate but exceedingly slow thermostat; and that the global cycles of carbon, nitrogen, sulfur, phosphorus and iron weave these processes into a coupled whole now driven beyond its Holocene boundaries. The decisive feature of this system is the asymmetry between fast amplifying feedbacks and slow stabilizing ones. Because the silicate-weathering negative feedback operates over hundreds of thousands of years while carbon–climate, permafrost, methane and ice–albedo feedbacks act within decades, the Earth system cannot chemically self-correct on any timescale relevant to human society. Stabilizing climate therefore depends not on natural chemical regulation but on reducing the chemical perturbation at its source, complemented where feasible by the deliberate acceleration of removal processes. A quantitative, cross-sphere understanding of these chemical feedbacks—their direction, magnitude, timescale and uncertainty—remains the indispensable scientific foundation for that effort.

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